Daily Problem: 3–Mar–2026

Problem Statement

Show that there are countably many Turing machines, but uncountably many subsets of Σ^*, and conclude that there exist languages that are not recursively enumerable.

Solution

1. Countably many Turing machines

A Turing machine T can be described by a finite set of states, an alphabet Σ, and a transition function δ, which is a mapping from state-symbol pairs to state-symbol-action triples. The key observation here is that the description of a Turing machine can be encoded as a finite binary string.

Step 1: Encoding a Turing machine.

- The finite set of states can be represented by a finite binary string (since each state corresponds to an integer, and the set of states is finite). - The tape alphabet Σ can also be encoded as a finite set of symbols, which can be represented by binary strings. - The transition function δ, which describes the behavior of the machine, can be encoded as a finite binary string (since it is a finite relation between the states and the symbols).

Thus, each Turing machine can be uniquely represented by a finite binary string.

Step 2: Countability of the set of Turing machines.

Since every Turing machine can be described by a finite binary string, and the set of all binary strings is countable, the set of all Turing machines is also countable.

Thus, there are countably many Turing machines.

2. Uncountably many subsets of Σ^*

Now, consider the set of all subsets of Σ^*, which are the possible languages over the alphabet Σ. The set of all subsets of Σ^* corresponds to the power set of Σ^*.

Step 1: Cardinality of Σ^*.

Σ^* is the set of all finite-length strings over the alphabet Σ. The size of Σ^* is countably infinite, as it includes all finite strings over a finite alphabet.

Step 2: Power set of a countable set.

The power set of a countable set S is the set of all subsets of S. It is known that the power set of a countable set is uncountably infinite. More precisely, if S is countably infinite, then the power set of S, denoted $\mathcal{P}(S)$, has the cardinality of the continuum, $2^{\aleph_0}$, which is uncountable.

Since Σ^* is countably infinite, the set of all subsets of Σ^*, or the set of all languages over Σ, is uncountable.

3. Conclusion: Existence of languages that are not recursively enumerable

The set of all languages over Σ is uncountable, but the set of all recursively enumerable languages (the set of languages that can be recognized by Turing machines) is countable, as we showed earlier.

Therefore, since the set of all languages is uncountable and the set of recursively enumerable languages is countable, there must exist languages that are not recursively enumerable.

In conclusion, there exist languages that cannot be recognized by any Turing machine, i.e., there are languages that are not recursively enumerable.